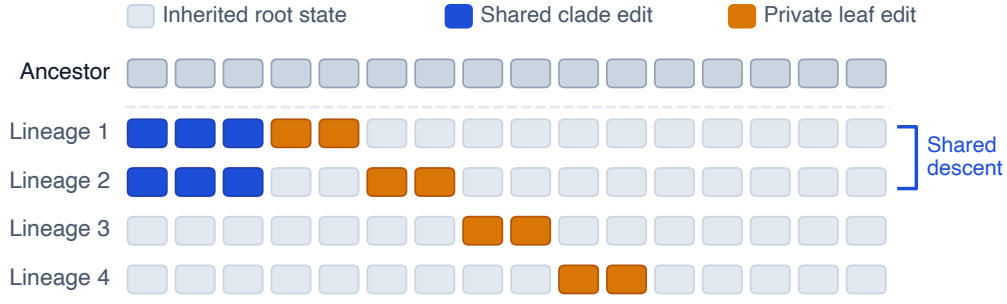
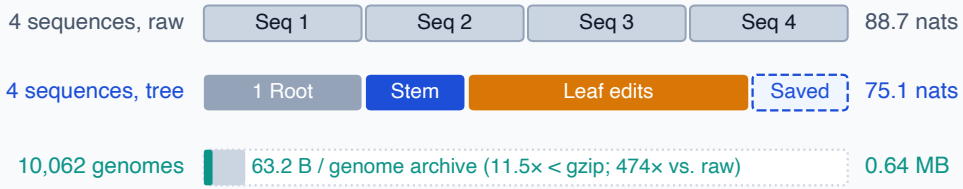
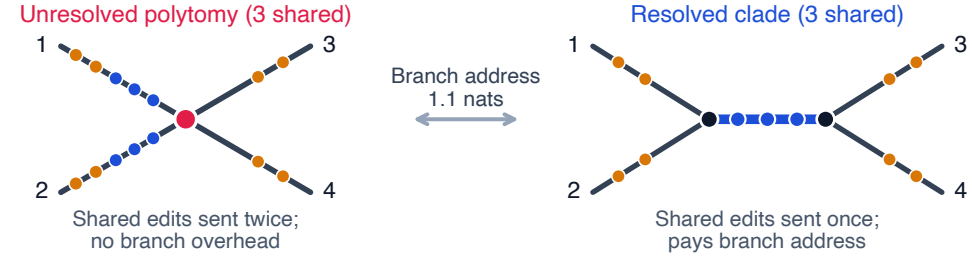
A Heredity as lossless compression



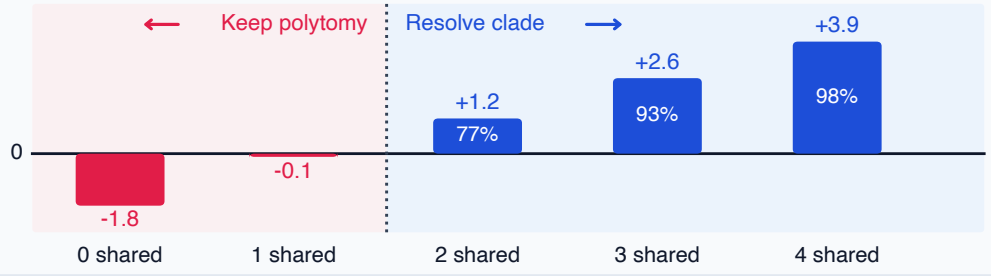
Physical description length required for exact lossless recovery



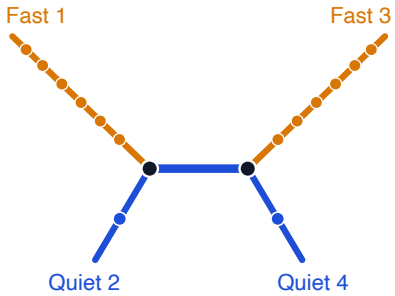
B When a branch earns its place: burst vs. split



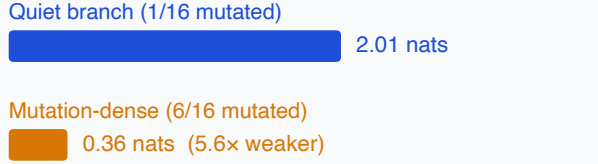
Net information saved by adding the branch (nats; % = posterior support)



C Uneven evolutionary rates and deep-time saturation

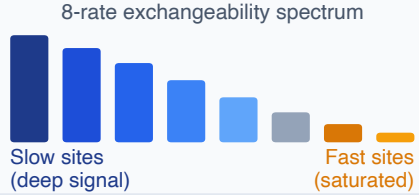


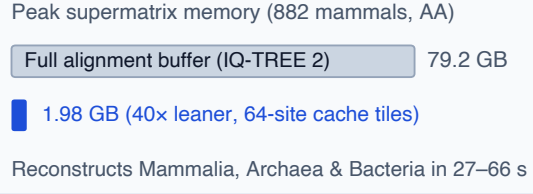
Positional information saved per shared match



Coincidental hits on fast stems save little length

Deep-time supermatrices: site-rate spectrum & cache-tiled Fisher MDL

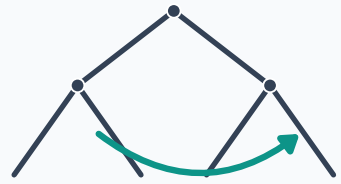




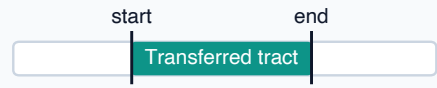
Reconstructs Mammalia, Archaea & Bacteria in 27–66 s

D Separating gene transfer from site convergence

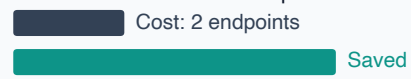
Contiguous block transfer



Lateral block transfer

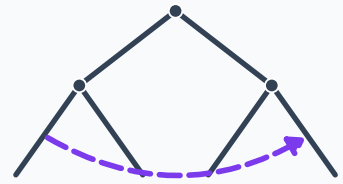


Link cost vs. shared edits explained



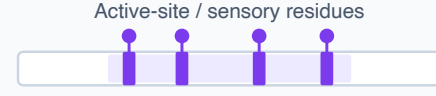
Recombination • HGT • Paralog conversion

Scattered site convergence



Parallel convergence

Active-site / sensory residues



Link cost vs. shared edits explained

Inner-ear hearing • Diet • Antibody CDRs